# Om Patel

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## Experience

**Quality & Automation Engineer Intern**, Relay Financial – Toronto, ON — May 2026 – Present
- Engineered a Linux VM environment for Cursor Cloud Agents, with rootless Docker, GPG/SHA-pinned supply chain, least-privilege sudoers, and dual Node toolchains, enabling AI agents to run Docker-based integration stacks unattended
- Reduced CI infrastructure usage by 6 node-hours/day and developer wait time by 5–7 hours/day by building a Datadog cost-attribution dashboard to optimize worker counts and migrate self-hosted GitHub Actions runner scale sets
- Increased test coverage from 0% to 55% for core in-house payments service by leading integration-test development, building its CI/CD test pipeline, workers, coverage reporting, and service mocks for AWS and third-party dependencies
- Cut regression suite runtime by 42% through test parallelization while scaling core banking automation to 300+ maintained tests, adding 125+ test cases across API, UI, Lambda, mobile, and web surfaces

**Software Test Intern**, FNZ – Toronto, ON — May 2025 – Aug 2025
- Reduced pending retest backlog by 50%+ and unblocked team specific development by identifying 30+ platform bugs and validating 75+ fixes across staging and user-test environments using SQL, RabbitMQ, API logs, and Postman
- Improved automated regression accuracy by 18% by debugging 45+ Cypress tests and converting manual Client Onboarding coverage into an end-to-end suite spanning 10+ workflows, edge cases, and core platform functionality
- Authored 6+ technical design documents for the FNZ-BMO Fund Classification workflow, translating Bloomberg and Morningstar data into client-facing dashboards, collateral calculations and downstream processing

**Data Analyst & Administrative Assistant**, University of Toronto – Toronto, ON — May 2024 – Aug 2024
- Presented PEY Co-op insights and improvement steps to 15+ director staff by analyzing 1,200+ survey responses
- Produced Annual and Seasonal Recruitment Reports by analyzing 2,000+ Excel student records to identify hiring trends

## Projects

**FPGA Optical Communications DSP** — Github | Aug 2026
- Designed a fixed-point optical receiver DSP chain (DC removal, 16-tap FIR, phase selection, PRBS sync) in SystemVerilog for a 650nm NRZ-OOK link on the Arty A7-100T, achieving 0.03% bit error rate across 5.6M bits at 10kbit/s
- Verified the receiver pipeline using bit-exact SystemVerilog testbenches and physical XADC captures, closing timing at 100MHz and tuning the 16-tap FIR to reduce signal noise by 45.8% while retaining 87.8% of the signal fundamental

**FPGA Sobel Vision Accelerator** — Github | July 2026
- Engineered a CPU-free FPGA vision accelerator on Arty A7-100T in SystemVerilog, integrating grayscale conversion and 3x3 Sobel edge detection, achieving 81,380 frames/s and 5.74x OpenCV throughput with 32 parallel lanes at 200 MHz
- Implemented custom Ethernet/UDP stack in RTL with MDIO/MII, ARP, IPv4, CRC/FCS and asynchronous FIFOs, streaming QVGA frames directly from FPGA to Streamlit dashboard at 30 FPS across 9,000 frames with zero transport errors

**WearAlert** — Github | Apr 2026
- Co-engineered a wireless fall-detection system for real-time caregiver alerts, fusing BNO055 motion and DPS310 barometric data across ESP32, ESP8266, and STM32 to achieve <100 ms end-to-end latency at 9–10 Hz
- Developed STM32 hardware interfaces and firmware in C, integrating interrupt-driven UART, 400 kHz I2C, GPIO/EXTI, and timer-triggered DAC/DMA to receive sensor telemetry and control OLED feedback and real-time audio alerts

**Radio Power Amplifier and Low-Pass Filter PCB** — Github | May 2025
- Designed and constructed a Class E power amplifier and 7-stage low-pass filter on a 2-layer Altium PCB for 16 MHz and 60V operation, achieving 2.7 W output, 30 dB gain, 76% power efficiency, and 1.69% total harmonic distortion
- Executed automated hardware unit testing using Python to interface with oscilloscope, DMM, and waveform generator

**Bread Crumb Trail - MakeUofT 2026 Hardware Hackathon Winner (Qualcomm Sponsor Prize)** — DevPost
- Built an Arduino UNO Q navigation/SOS system with GNSS, cellular connectivity, and real-time breadcrumb tracking

## Skills

**Languages**: C, C++, Python, SystemVerilog, Verilog, RV32 Assembly, JavaScript, SQL
**Design & Development**: Vivado, Quartus, ModelSim, Altium, LTspice, MATLAB, SUE SoC, MMI MAX
**Hardware**: Arty A7-100T (Artix-7), DE1-SoC, STM32 (Cortex-M4), ESP32, Arduino, OV7670, Oscilloscopes, Soldering

## Education

**University of Toronto –** BASc in Electrical Engineering — **Expected May 2028**